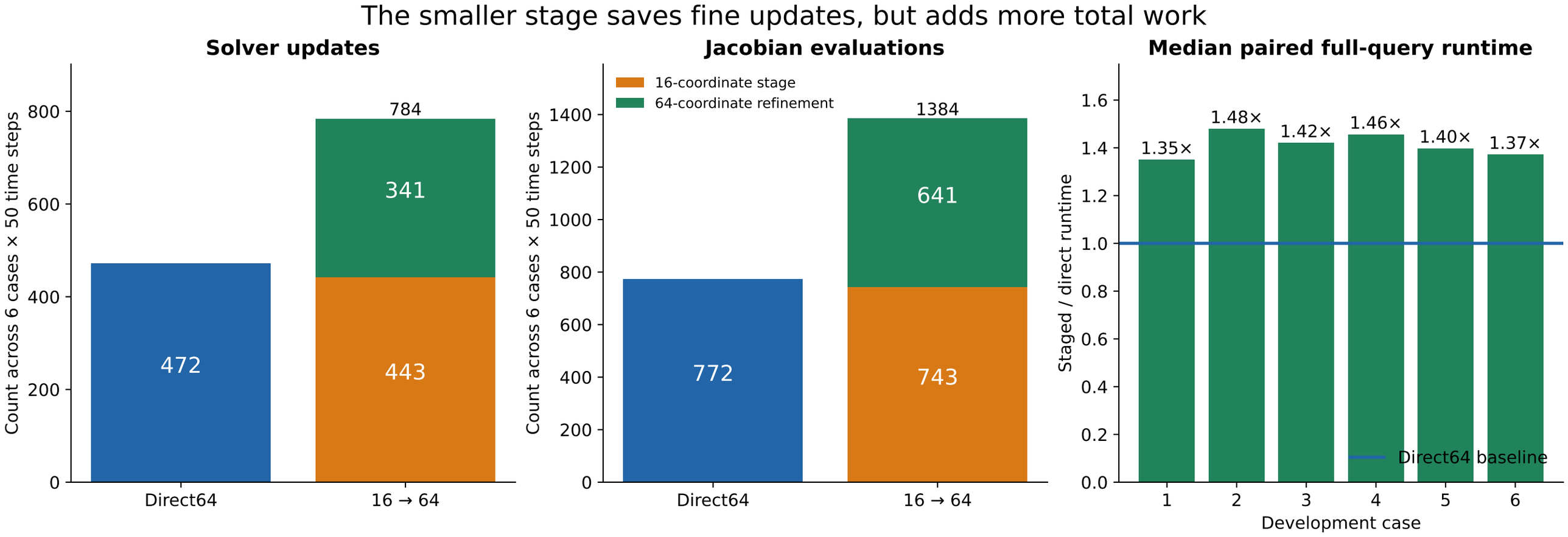
The smaller stage saves fine updates, but adds more total work
Solver updates
Count across 6 cases × 50 time steps
0
200
400
600
800
472
784
341
443
Direct64
16 → 64
Jacobian evaluations
16-coordinate stage
64-coordinate refinement
Count across 6 cases × 50 time steps
0
200
400
600
800
1000
1200
1400
772
1384
641
743
Direct64
16 → 64
Median paired full-query runtime
Staged / direct runtime
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.35×
1.48×
1.42×
1.46×
1.40×
1.37×
Direct64 baseline
1
2
3
4
5
6
Development case